

ABSTRACT

Medical Specialty Classification using TF-IDF and Ensemble Machine Learning

Manually categorizing medical transcription reports into clinical specialties is time-consuming, labor-intensive, and prone to human error. This project proposes the development of an automated Medical Specialty Classification system utilizing Natural Language Processing (NLP) and Machine Learning to streamline document organization.

Preprocessing will involve lowercasing, punctuation removal, tokenization, stop-word removal, and lemmatization. Term Frequency–Inverse Document Frequency (TF-IDF) will be used to convert clinical text into numerical vectors. Multiple machine learning algorithms, including Logistic Regression, Support Vector Machine (SVM), Random Forest, and Multinomial Naïve Bayes, will be trained and evaluated. A Voting Ensemble Classifier will then combine these predictions to maximize classification performance.

To ensure practical accessibility, a Flask-based web application will be developed, enabling users to submit medical transcription reports and instantly obtain the predicted medical specialty. The proposed system aims to improve the organization and retrieval of clinical documents, reduce manual classification effort, and demonstrate the effectiveness of TF-IDF with ensemble machine learning for accurate medical specialty classification.

Dataset Source: Medical Transcriptions (MTSamples) Dataset, Kaggle

<https://www.kaggle.com/datasets/tboyle10/medicaltranscriptions>

Technology Stack:

Frontend: HTML, CSS, Bootstrap

Backend: Python, Flask

Machine Learning & NLP: TF-IDF, Logistic Regression, Support Vector Machine (SVM), Random Forest, Multinomial Naïve Bayes, Voting Ensemble Classifier

Libraries: Scikit-learn, Pandas, NumPy, NLTK, Joblib

IDE & Tools: VS Code, Google Colab, Jupyter Notebook

Database: Not required. The Medical Transcriptions dataset is maintained in CSV format and serves as the primary data source.

References:

- [1] A. E. Blanchard et al., “A Keyword-Enhanced Approach to Handle Class Imbalance in Clinical Text Classification,” *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 6, pp. 2796–2803, Jun. 2022. (<https://doi.org/10.1109/JBHI.2022.3141976>)
- [2] A. M. Omar, M. Elhoseny, and A. M. Hassanien, “Clinical Text Classification with Word Representation Features and Machine Learning Algorithms,” *International Journal of Online and Biomedical Engineering (iJOE)*, 2023. (<https://online-journals.org/index.php/i-joe/article/view/36099>)
- [3] S. U. Amin, M. Alsulaiman, G. Muhammad, M. A. Mekhtiche, and M. S. Hossain, “Medical Specialty Classification Based on Semiadversarial Data Augmentation,” *IEEE Access*, 2023. (<https://pubmed.ncbi.nlm.nih.gov/37881209/>)

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